

*“Al Jazeera 360 Platform – Data Analyst
Assessment”*

Prepared by Wael, September 2025

Task 1: Total Views by Source (Brand)

1. Objective

Calculate the number of views per content source (brand). A “view” is defined as one unique combination of user, content, and calendar day, so repeated plays of the same content by the same user on the same day are only counted once.

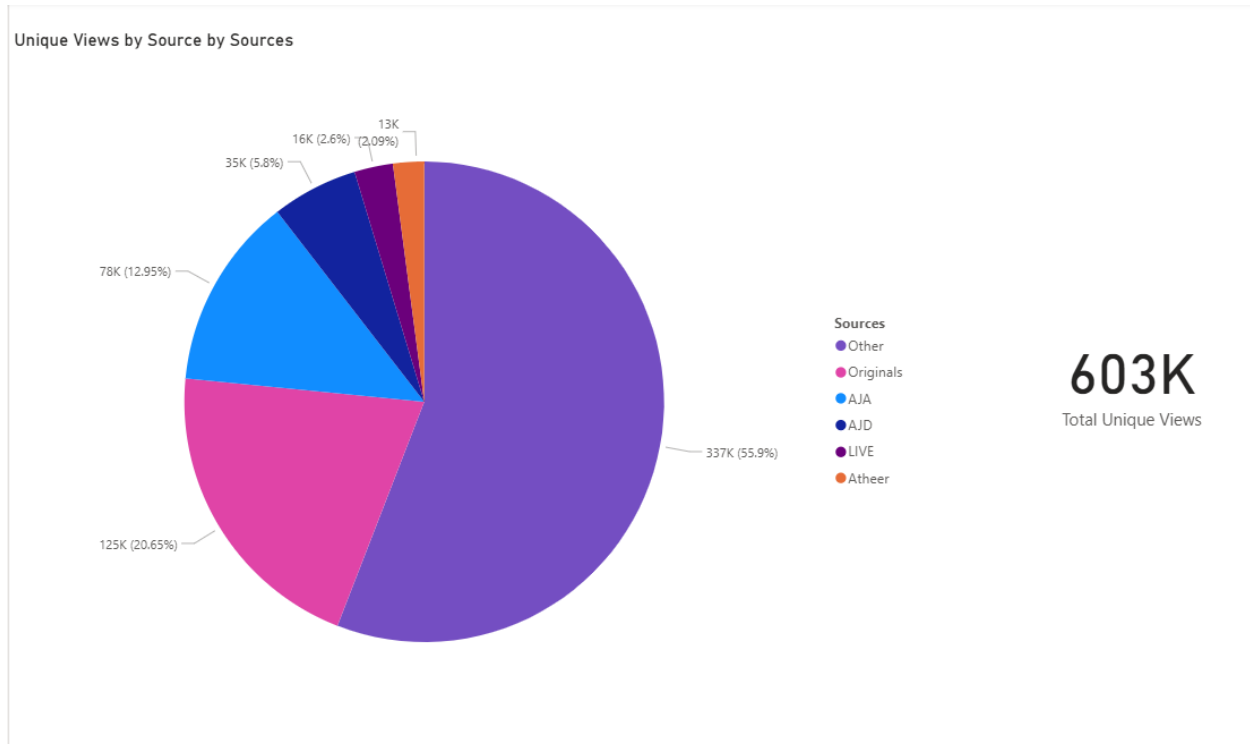
2. Method

- Joined viewrship (watch events) with catalog (content metadata) using CONTENT_ID = vod_dve_id.
- Classified content into sources by external_id prefix:
 - PGA → AJA (Al Jazeera Arabic)
 - PGD → AJD (Al Jazeera Documentary)
 - PGV → Atheer (Podcasts)
 - PGL → LIVE (Live Channels)
 - PGC → Originals (Original productions/commissions)
 - Otherwise → Other
- Built a unique view key = CUSTOMER_EXID + CONTENT_ID + DATE(STARTED_AT).
- Counted distinct keys per brand.
- Sorted results in descending order.

3. Results

source	unique_views
Other	337,126
Originals	124,570
AJA	78,118
AJD	34,979
LIVE	15,696
Atheer	12,610

4. Visualization (Power BI)



5. Assumptions

- A “unique_view” = (user, content, day).
- Content IDs with no matching catalog entry were grouped under Other.
- Blank or missing external_id values were also included in Other.

6. Findings

- The Other category is the largest by far (over 50% of views). This likely reflects missing or unclassified external_id values.
- Originals is the top classified source (~125k unique views), showing strong audience interest in commissioned productions.
- AJA and AJD also have significant viewership, confirming the relevance of Arabic and Documentary content.
- LIVE and Atheer are relatively small, with modest engagement compared to the rest.
- This suggests metadata quality is an issue (too many "Other") but also highlights audience demand for Originals and core Al Jazeera Arabic content.

7. Recommendations

- **Fix metadata quality:** Clean and standardize external_id values in the catalog to shrink the Other bucket.
- **Lean into Originals:** Since Originals are attracting the most attention, continue investing in this type of content.
- **Maintain core brands:** AJA and AJD remain important pillars and should be supported, but Originals should be the growth focus.
- **Monitor smaller sources:** LIVE and Atheer could be opportunities in the future, but for now avoid over-investment until audience demand is clearer.

Task 2: Average Progress per Video

1. Objective

Measure how far viewers get through each video on average (percentage of video watched).

2. Method

- Used PROGRESS (seconds watched) and CONTENT_DURATION (total seconds).
- Progress per video % = $(\text{PROGRESS} / \text{CONTENT_DURATION}) \times 100$, capped at 100% as some progress might exceed 100% (repeated watch)
- Grouped progress as bins of 5% on Power BI.
- Ignored rows with missing/zero duration.
- Averaged the progress % for each video (CONTENT_ID).
- Joined with catalog table to show video titles.

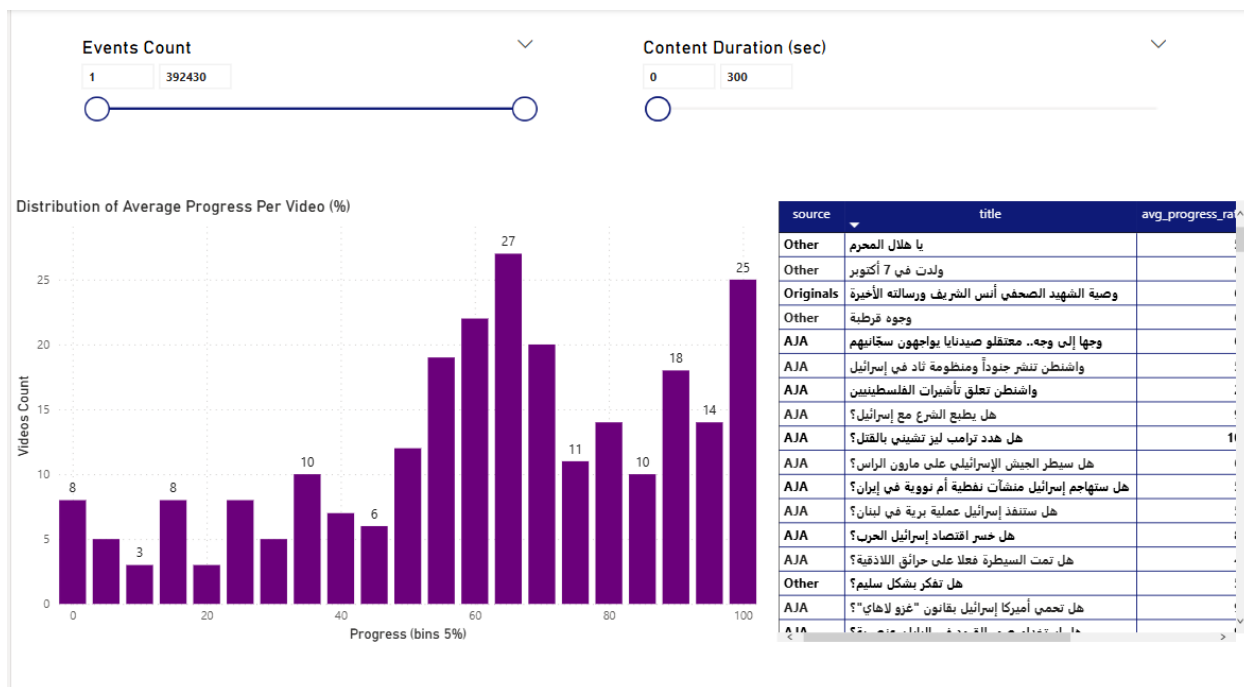
3. Results

CONTENT_ID	source	title	events_count	content_dureation_sec	avg_progress_ratio
851924	AJA	مهام الأجهزة الأمنية بين القمع والحماية	1	2830	100.0
851018	AJA	الكنيسة الأنجليكانية في كندا والسكان الأصليون	1	1682	100.0
851017	AJA	الاتجار بالفتيات في إيطاليا	1	1646	100.0
851016	AJA	تفاصيل محاولة اغتيال البابا في الفلبين	2	1728	100.0
851015	AJA	تفاصيل مقتل الأسرة المالكة النيبالية	1	1678	100.0
851013	AJA	ناورو: من الازدهار إلى الأزمات	1	1680	100.0
851012	AJA	ماليزيا بين قانون الطوارئ والديمقراطية	2	1689	100.0
851011	AJA	العلمانية التركية والحرب ضد التدين	1	1549	100.0
851010	AJA	نضال المرأة النيوزيلندية	3	1683	100.0

631006	AJA	صناعة القرار في إسرائيل	1	2845	0.0
630829	Other	سر النغمة الغامضة	1	982	0.0
630792	AJD	متمردان بين حريين	1	1642	0.0
630595	Other	علاج رخيص مخفي لمرض السرطان	1	1007	0.0
630521	Other	الأواني المنزلية السامة	1	531	0.0
630077	AJD	سيرة الزير سالم	1	800	0.0
631006	AJA	صناعة القرار في إسرائيل	1	2845	0.0

(Full list data shown in Power BI dashboard)

4. Visualization (Power BI)



5. Assumptions

- PROGRESS represents seconds watched.
- CONTENT_DURATION is total video length in seconds.
- Each per-event ratio is capped at 100%.
- Rows with missing/zero duration excluded.

6. Findings

- Many short clips and promos achieve 70 - 100% average progress.
- Long-form videos (documentaries & full-length programs) generally show lower averages, but a few still achieve strong completion.
- Zero/near-zero averages are mostly due to PROGRESS = 0 events (opened but not watched) or very few total plays.

7. Recommendations

- Maintain filters (content duration, events counts) in dashboards to reduce noise from low-sample titles.
- Investigate zero-progress videos: could indicate technical playback issues or low engagement starts.
- For long videos, improve the hooks in the first minutes and use chapters to keep viewers engaged.
- Check videos with consistently high progress %, since they perform better at keeping the audience engaged.

Task 3: Average Completion Rate

1. Objective

Measure how often viewers finish videos, grouped by source. A view is considered “completed” if at least 90% of the video was watched.

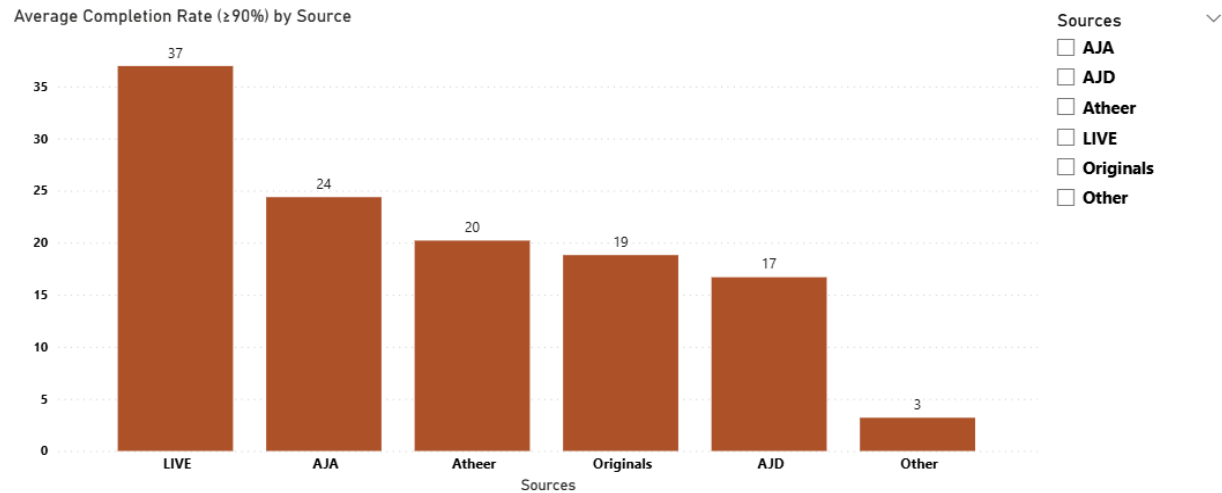
2. Method

- Filtered out rows where CONTENT_DURATION was missing or 0.
- Defined completion = $(\text{PROGRESS} \div \text{CONTENT_DURATION}) \times 100 \geq 90$.
- Counted total valid events per brand, and how many reached completion.
- Computed completion_rate_pct = $\text{completed_events} \div \text{events_count} \times 100$.
- Classified sources by external_id prefix:

3. Results

source	events_count	completed_evets	completed_rate_pct
LIVE	24810	9171	36.9649334945586
AJA	107975	26317	24.3732345450336
Atheer	21888	4422	20.202850877193
Originals	179636	33789	18.8097040682269
AJD	46445	7748	16.6820971041016
Other	624325	19930	3.19224762743763

4. Visualization (Power BI)



5. Assumptions

- PROGRESS = seconds watched.
- CONTENT_DURATION = total length in seconds.
- A completion is defined at **90% watched**.
- Excluded events with missing or zero duration.
- If external_id missing, grouped as “Other”.

6. Findings

- LIVE content has the strongest completion (~37%), which points to highly engaged audiences during live sessions.
- AJA sits in the middle with around 24% completion, performing better than AJD and Originals.
- Atheer and Originals are on the lower side (~20%), with AJD slightly below that.
- The “Other” category is very low (~3%), which likely reflects metadata gaps rather than true performance.

7. Recommendations

- **Clean up metadata:** Reduce reliance on “Other”, this will give a clearer view of performance by source.
- **Investigate LIVE:** Understand what drives its relatively high completion, perhaps shorter formats or more engaged audiences.
- **Boost long-form titles:** Originals and AJD have lower completion; consider breaking long programs into shorter segments or use chapters to keep viewers engaged.
- **Monitor podcasts (Atheer):** Engagement is weak; could need format tweaks or better promotion.

Task 4: Most-watched series (unique viewers)

1. Objective

Find the most-watched series by unique viewers, and identify which sources they belong to.

2. Method

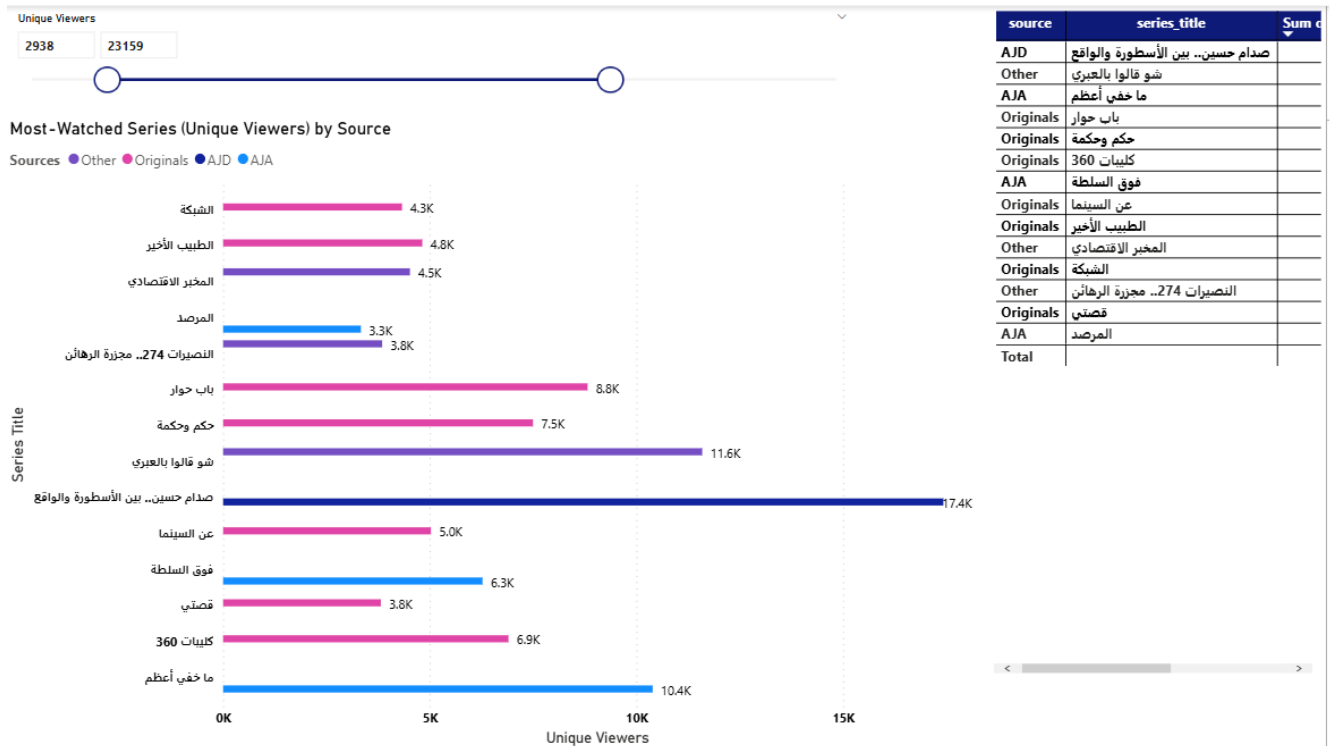
- Grouped viewership by SERIES_ID.
- Counted distinct CUSTOMER_EXID per series (unique viewers).
- Counted all rows per series (total views).
- Excluded unavailable SERIES_ID
- Joined to catalog for brand classification.
- Sorted by unique viewers (descending).

3. Results

sources	SERIES_ID	series_title	unique_viewers	total_views
Originals	3409	خيال أم جد؟	32317	77656
AJD	3125	صدام حسين.. بين الأسطورة والواقع	17427	27825
Other	2678	شو قالوا بالعبري	11592	17707
AJA	2207	ما خفي أعظم	10393	19157
Originals	2407	باب حوار	8818	20838
Originals	2276	حكم وحكمة	7498	13855
Originals	3433	كليات 360	6904	7939
LIVE	3028	هيل وزعفران	4	4
Other	3054	رمضان من	4	9
Other	3092	معالم إسلامية	4	6
AJA	2392	الهند البحث عن الروح	3	7
AJA	2409	جلادون وضحايا	3	4
AJA	2547	حكاية الوادي	3	6

(Full list data shown in Power BI dashboard)

4. Visualization (Power BI)



5. Assumptions

- Unique viewer = distinct CUSTOMER_EXID per series regardless of how many episodes or replays they watched.
- Grouped using SERIES_ID and SERIES_TITLE from viewership data.
- If SERIES_TITLE was missing, replace with "Unknown Series".
- If external_id missing, grouped as "Other".

6. Findings

- Originals lead the rankings: multiple series (خيال أم جد؟، باب حوار، حكم وحكمة، عن السينما) show strong unique audiences.
- Flagship series from AJD and AJA (like صدام حسين.. بين الأسطورة والواقع and ما خفي أعظم) also rank high, confirming the importance of these core sources.
- Other bucket includes popular series (e.g. شو قالوا بالعبري، المخبر الاقتصادي). This suggests metadata issues (series not properly classified).
- Some series have very high total views vs. unique viewers (heavy replay, loyal audiences). Others show broad reach but fewer repeat plays.

7. Recommendations

- **Prioritize Originals:** Keep investing in Originals: they consistently top the list (e.g. عن السينما)
- **Back AJA & AJD flagships:** Support core series from AJA and AJD, as they remain important to the audience (e.g. ما خفي أعظم)
- **Fix metadata:** Clean up “Other” to surface popular shows (e.g. المخبر الاقتصادي)
- **Segment by loyalty vs. reach:**
 - Heavy replay: target loyalty/monetization (e.g. باب حوار)
 - Broad-reach: focus on awareness campaigns (e.g. خيال أم جد؟)
- **Catalog review:** Many with 1-2 viewers may need promotion, reclassification, or retirement.

Task 5: New vs. Old Content

1. Objective

Analyze how viewership differs between newer and older titles, and compare across sources.

2. Method

- Used catalog.created_at to classify content into New and Old.
- Year = 2025 (found via pattern '/2025 ') AND month in (7,8,9)
- Joined viewership (watch events) with catalog on CONTENT_ID = vod_dve_id.
- Classified content into sources by external_id prefix.
- Unique viewers = distinct CUSTOMER_EXID per source and age bucket.
- Total views = all watch events (including repeats).
- Aggregated results into a two-way split (content_age × source).
- Included rows where created_at available

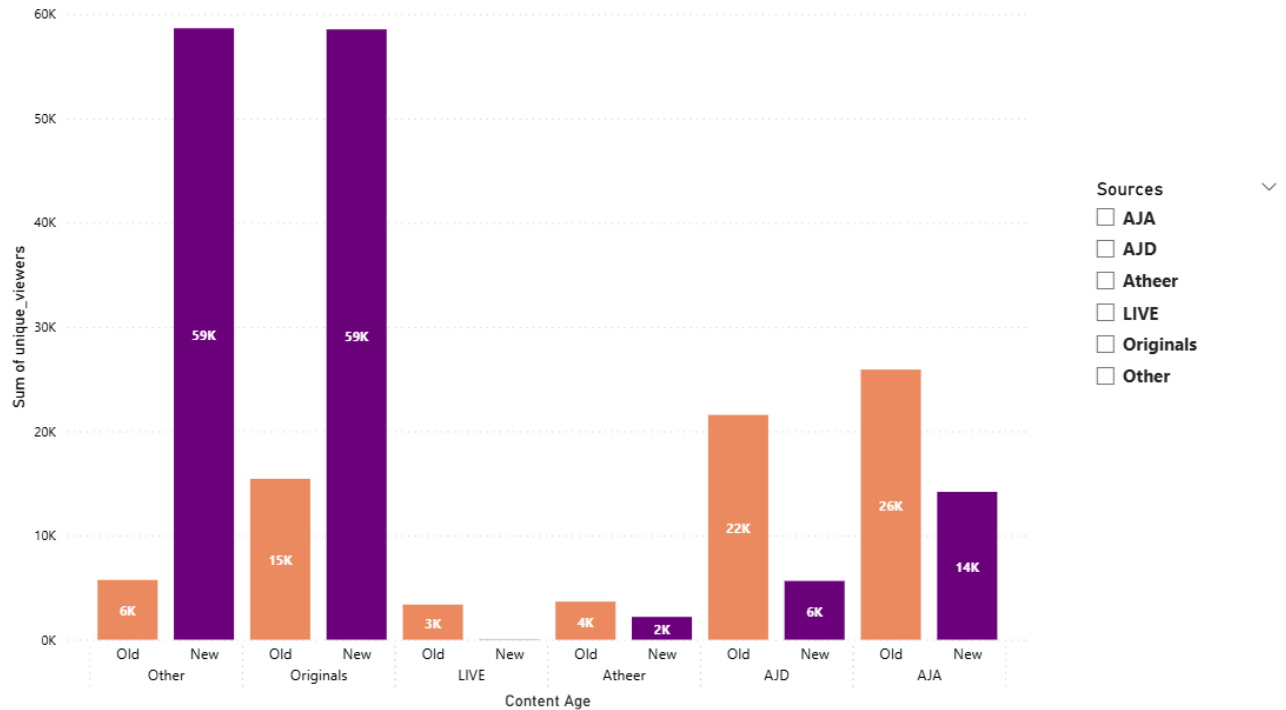
3. Results

content_age	source	unique_viewers	total_views
Old	AJA	25915	75732
Old	AJD	21567	37626
Old	Originals	15432	38532
Old	Other	5732	12620
Old	Atheer	3654	17016
Old	LIVE	3367	24772
New	Other	58654	113511
New	Originals	58552	141104
New	AJA	14179	32243
New	AJD	5637	8819
New	Atheer	2186	4872
New	LIVE	27	38

4. Visualization (Power BI)

Unique viewers by Sources & Content Age

Content Age ● Old ● New



5. Assumptions

- New content = titles created in Jul-Sep 2025 (last 3 months of dataset).
- Old content = titles created before July 2025.
- If external_id missing, grouped as "Other".

6. Findings

- New Originals dominate (~58.5k unique viewers, ~141k total views), showing audience appetite for fresh, commissioned productions.
- New Other content also scored high (~58k unique viewers), which signals metadata issues, popular new titles aren't being properly classified.
- Old AJA and AJD content still perform strongly (~25k and ~21k unique viewers), proving the long-term value of the archive.
- Atheer podcasts perform better in Old (~3.6k) vs. (~2.2k) New, suggesting a stable but limited audience.
- LIVE appears almost entirely Old, since the New/Old split doesn't fit continuous streams

7. Recommendations

- **Prioritize New Originals:** they're proving to be the biggest drivers of audience growth.
- **Clean up metadata for New content:** reduce reliance on "Other" so performance attribution is accurate.
- **Promote archive hits (Old AJA/AJD):** Highlight strong historical content in user suggestions, as they still attract solid audiences.
- **Rethink Atheer strategy:** invest in improving podcast engagement through promotion or format changes.
- **Measure LIVE separately:** use metrics like average watch time or completion instead of New/Old classification.

Task 6: Repeat Viewers (overall and by source)

1. Objective

Measure how many users return on a different calendar day within the dataset, and compare repeat behavior by source (AJA, AJD, Originals, LIVE, Atheer, Other).

2. Method

- Took every watch event and reduced the timestamp to a date (e.g., “9/3/2025”), so sessions on the same day don’t inflate the count.
- Built a list of (user, date, content) to preserve the link to source via the catalog.
- For each user × source, counted how many distinct days they were active:
 - Repeat user for that source = active on > 1 day.
- Aggregated per source:
 - total_users = distinct users who watched that source,
 - repeat_users = users with > 1 active day for that source,
 - repeat_share_pct = repeat_users / total_users × 100.

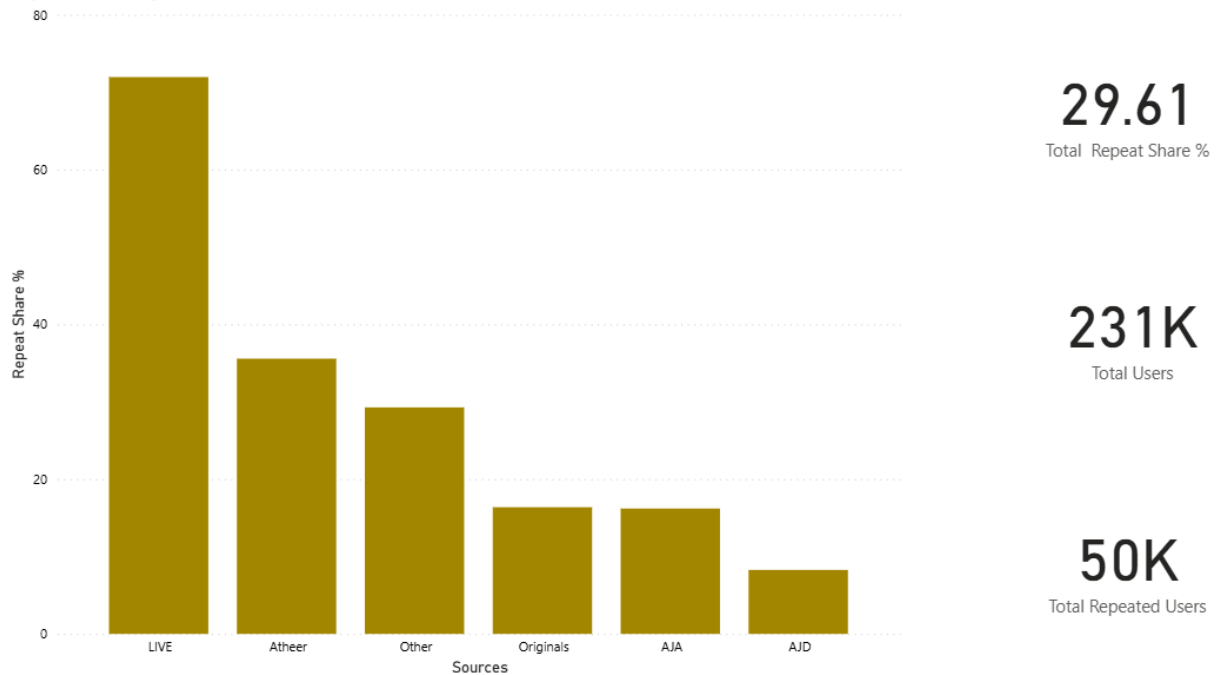
3. Results

source	total_users	repeat_users	repeat_share_pct
LIVE	3,387	2,437	71.95%
Atheer	5,302	1,886	35.57%
Other	93,243	27,290	29.27%
Originals	66,693	10,918	16.37%
AJA	35,737	5,790	16.20%
AJD	26,204	2,166	8.27%

(Overall Repeated Share %: 29.61%)

4. Visualization (Power BI)

Repeat Share % by Sources



5. Assumptions

- Repeat user = the same CUSTOMER_EXID active on different calendar days (local date strings), not just multiple sessions in one day.
- Repeat is calculated per source; a user can be repeat in one source and one-time in another.
- Some items fall under “Other” due to missing/unclean external_id.
- Dataset timeframe defines the observation window (no lifetime behavior).

6. Findings

- Overall, out of 163,139 total users, 51,200 were repeat viewers, giving a repeat share of 31.38%. Roughly 1 in 3 users came back on a different day.
- LIVE has the highest repeat share (~72%), confirming that live channels are highly habit-forming.
- Atheer podcasts also perform relatively well (~36%), likely due to episodic listening behavior.
- “Other” sits at 29% but is inflated by metadata issues; many users here probably belong to other brands.

- Originals and AJA both have lower repeat shares (~16%), meaning strong audiences but weaker return visits.
- AJD is lowest at 8%, suggesting one-time, event-driven consumption dominates.

7. Recommendations

- **Use LIVE to keep viewers coming back:** Make live schedules easy to find, and link live events to VOD highlights or follow-ups so the engagement carries over.
- **Build on the success of episodic formats:** Atheer shows that series-style releases keep people returning, consider more weekly or serialized drops for Originals and AJA.
- **Make it easier to rewatch VOD (Originals/AJA/AJD):**
 - Add “*Up next*” and “*Continue watching*” rails.
 - Send reminders when new episodes or related content are out.
 - Include chapters and quick recaps so viewers can jump back in easily.
 - Clean up metadata so repeat views are tied to the right brand.
- **Strengthen AJD retention:** Test shorter edits, mini-series packaging, and punchier openings, and promote “what’s next” directly in the player.

Task 7: Top-Viewed Content List (Episodes with Thumbnails)

1. Objective

Identify the most-watched individual episodes, including their series, source and thumbnail, to understand which specific titles resonate most with audiences.

2. Method

- Joined viewership with catalog on CONTENT_ID = vod_dve_id.
- Defined a unique view as (user + content + calendar day).
- Grouped by episode title, series title, thumbnail, and source
- Excluded null or blank titles to keep the list clean.
- Ranked results by unique_views in descending order, limited to Top 20 episodes.

3. Results

source	title	series_title	thumbnail_url	unique_views
Other	كنت أعرف الخميني		https://dve-images.imggaming.com/original/p/2025/07/05/K5jU0seHZCkxbx4HGH4JBLI0vU5jE4oN-1751705010024.jpg	26548
Other	برومو خيال أم جد؟ الخواجه "ة" وعنق الزجاجة		https://dve-images.imggaming.com/original/p/2025/08/31/5uMu9mzMTifPJPIQ2STqDgb4tUFdfQgX-1756629413310.jpg	22196
AJD	صدام حسين.. بين الأسطورة والواقع - الجزء 1	صدام حسين.. بين الأسطورة والواقع	https://dve-images.imggaming.com/original/p/2025/04/04/tHsPGr2daorsBcInVJkBXpU9aq9AO5a4-1743780636452.jpg	17537
Origins	الخواجه "ة" وعنق الزجاجة	خيال أم جد؟	https://dve-images.imggaming.com/original/p/2025/08/29/W7npVQO5Z9tRBKwpTMHIX8qP6A0oqacJ-1756479378132.jpg	14868
Origins	أنهار العنصرة	خيال أم جد؟	https://dve-images.imggaming.com/original/p/2025/08/22/lWfRyulego783g2gQPvX6fYrk6BUxAcr-1755876058656.jpg	13104

Originals	من الطالق؟	خيال أم جد؟	https://dve-images.imggaming.com/original/p/2025/08/01/cjrdQcs2xDufLsColqx6nev81ohCuP0R-1754036358602.jpg	8780
Originals	زمن التكبيس	خيال أم جد؟	https://dve-images.imggaming.com/original/p/2025/08/08/o8GBZjfrCMD9gKPNJwglxms3PL6F10YT-1754642297825.jpg	7526
AJA	أربعون دقيقة	ما خفي أعظم	https://dve-images.imggaming.com/original/p/2024/11/10/Mw7buA7Hdjz9W8ii7EmNLFRpxBMM0dV0-1731221469669.jpg	7358

(Full Top 20 list with thumbnails shown in Power BI dashboard.)

4. Visualization (Power BI)

thumbnail_url	episode	series_title	source	unique_views
https://dve-images.imggaming.com/original/p/2025/07/05/K5jU0seHZCkxbx4HG4JBL0vU5jE4oN-1751705010024.jpg	كنت أعرف الخميني		Other	26548
https://dve-images.imggaming.com/original/p/2025/08/31/5uMu9mzMTifPJPIQ2STqDgb4tUfDfQgX-1756629413310.jpg	برومو خيال أم جد؟ "الخواجة" وعنق الرجاجة		Other	22196
https://dve-images.imggaming.com/original/p/2025/04/04/tHsPGr2daorsBclnVJkBXpU9aq9AO5a4-1743780636452.jpg	صدام حسين.. بين الأسطورة والواقع - الجزء 1	صدام حسين.. بين الأسطورة والواقع	AJD	17537
https://dve-images.imggaming.com/original/p/2025/08/29/W7npVQO5Z9tRBKwpTMHIX8qP6A0oqacJ-1756479378132.jpg	"الخواجة" وعنق الرجاجة	خيال أم جد؟	Originals	14868
https://dve-images.imggaming.com/original/p/2025/08/22/IWfRyulego783g2gQPvX6fYrk6BUxAcr-1755876058656.jpg	أنهار العنصرة	خيال أم جد؟	Originals	13104
https://dve-images.imggaming.com/original/p/2025/08/01/cjrdQcs2xDufLsColqx6nev81ohCuP0R-1754036358602.jpg	من الطالق؟	خيال أم جد؟	Originals	8780
https://dve-images.imggaming.com/original/p/2025/08/01/cjrdQcs2xDufLsColqx6nev81ohCuP0R-1754036358602.jpg	زمن التكبيس	خيال أم جد؟	Originals	7526
Total				170557

Sources

Select all
AJA
AJD
Originals
Other

Series Title

ما خفي أعظم
قصتي
صدام حسين.. بين الأسطورة والواقع
خيال أم جد؟
باب حوار

كليات 360
عن السينما
شو قالوا بالعيري
حكم وحكمة
الطبيب الأخير

5. Assumptions

- Unique views = (CUSTOMER_EXID, CONTENT_ID, date) as before.
- Only titles with non-null, non-blank values are included.
- Some entries in "Other" are due to incomplete or missing external_id classification.
- Thumbnails are taken directly from catalog thumbnail_url.

6. Findings

- Originals dominate the top list, particularly (عن السينما and حكم وحكمة, باب حوار, خيال أم جد؟).
- Core AJD and AJA titles are also strong performers (ما خفي أعظم, صدام حسين. بين الأسطورة والواقع).
- “Other” bucket includes popular hits (المخبز الاقتصادي, شو قالوا بالعبري), showing metadata issues.
- Several series have multiple episodes in the Top 20, showing strong loyalty and repeat interest within those franchises.

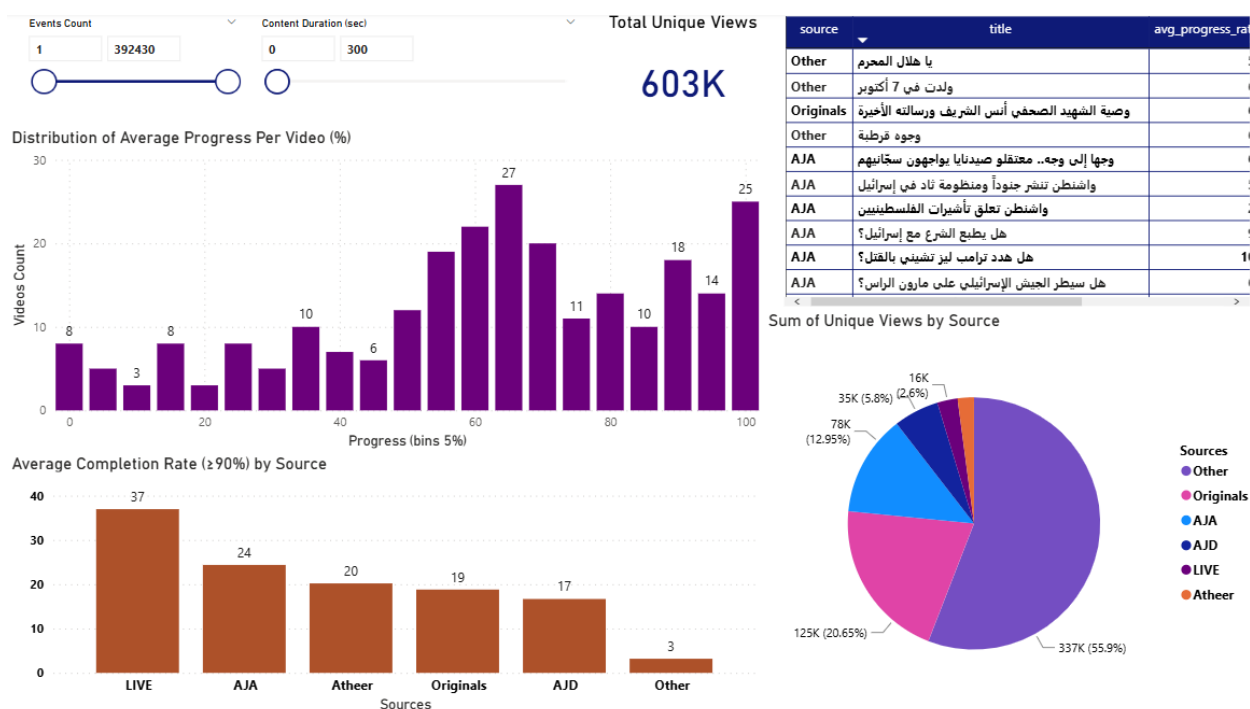
7. Recommendations

- **Promote Originals strongly:** They consistently lead the charts and deserve priority placement in campaigns and home pages.
- **Sustain core AJA/AJD:** These hits remain central to audience engagement and should keep strong promotional support.
- **Fix metadata quality:** ensure popular series like شو قالوا بالعبري are properly attributed not hidden under “Other.”
- **Use strong visuals:** Feature thumbnails prominently in app and website discovery to spotlight top-performing content.
- **Learn from repeat hits:** Titles with multiple episodes in the Top 20 can guide future production and scheduling.

Final Conclusion & Recommendations

1. Conclusion

- Originals are the strongest audience driver. New titles and series consistently top the charts, showing that commissioned productions really connect with viewers.
- Core brands (AJA and AJD) still matter. Flagship shows like **صدام حسين بين** and **ما خفي أعظم** underline the lasting appeal of archive and documentary content. However, their repeat engagement is weaker compared to LIVE and episodic formats.
- LIVE builds loyalty. With ~72% completion and repeat rates, LIVE content shows strong habit-forming behavior, even if its overall reach is smaller than VOD.
- Podcasts (Atheer) show promise: Audiences are modest, but the episodic format keeps people coming back more than most VOD categories, a sign that with the right tweaks and promotion, it could grow.
- The “Other” bucket is too large, often containing popular shows (**إشو قالوا بالعبري**, **المخير**) (الاقتصادي) that should be properly attributed. Cleaning this up would give a truer view of brand performance.
- User engagement patterns show that short clips and promos drive very high progress and completion, while long-form programs struggle. Overall, only ~30% of users come back, pointing to big opportunities to create stickier, more repeatable experiences.



2. Recommendations

- Clean up and standardize external_id values to shrink the “Other” bucket
- Make sure popular titles like *شو قالوا بالعبري* are tied to the right brand so performance tracking is accurate.
- Originals remain the biggest audience magnet, keep investing heavily.
- Give them prime visibility on the homepage, in campaigns, and in recommendations.
- Use proven series like *باب حوار*, *خيال أم جد؟*, and *عن السينما* as benchmark for future productions.
- Maintain strong promotion for (AJA & AJD) flagship titles, as they remain central to Al Jazeera’s identity.
- Experiment with shorter edits, serialized packaging, and “next up” nudges to boost repeat engagement.
- Maximize the loyalty of LIVE audiences by making schedules easy to find and linking live events to VOD highlights.
- Use LIVE as a gateway, pushing viewers to replays, related Originals, or AJD docs.
- Add chapters, recaps, and ‘continue watching’ rails to encourage viewers to finish long programs.
- Strengthen hooks in the first minutes of long-form content to reduce early drop-offs.
- Episodic content (like Atheer podcasts and Originals series) shows higher repeat behavior, explore more weekly/serialized releases.
- Use reminders and notifications for new drops to build habits.
- Separate loyalty shows (e.g., *باب حوار*) from broad-reach shows (e.g., *خيال أم جد؟*) and promote accordingly:
Loyalty-driven -> target monetization and subscriber retention.
Broad-reach -> use for brand awareness and audience expansion.
- Keep highlighting strong archive content (AJA/AJD older titles) in recommendations rails in order to capitalize on proven long-term interest.

The AJ360 platform has clear strengths in Originals and core Al Jazeera brands, alongside a highly loyal LIVE audience.

However, metadata issues and weaker repeat engagement in long-form VOD limit the platform's potential.

By cleaning data, focusing on Originals, and improving retention mechanics, AJ360 can expand reach and deepen loyalty, aligning content performance more closely with audience growth and engagement goals.

Appendix

```
1 -- Task 1: Total views by source (brand)
2
3 -- Select the source name and the number of views for each source
4 SELECT
5     -- Classify external_id prefixes to readable source names
6     CASE
7         WHEN c.external_id LIKE 'PGA%' THEN 'AJA'           -- If it starts with PGA → Al Jazeera Arabic
8         WHEN c.external_id LIKE 'PGD%' THEN 'AJD'           -- If it starts with PGD → Al Jazeera Documentary
9         WHEN c.external_id LIKE 'PGV%' THEN 'Atheer'         -- If it starts with PGV → Atheer (Podcasts)
10        WHEN c.external_id LIKE 'PGL%' THEN 'LIVE'           -- If it starts with PGL → Live channels
11        WHEN c.external_id LIKE 'PGC%' THEN 'Originals'       -- If it starts with PGC → Originals / Commissions
12        ELSE 'Other'                                          -- Everything else or missing external_id
13    END AS source, -- name it as source
14
15    -- Count view units for each source
16    -- A unique view = 1 user watching 1 video on 1 day.
17    -- So if the same user watches the same video 3 times in the same day → count = 1.
18    -- We build this by combining CUSTOMER_EXID + CONTENT_ID + the date part of STARTED_AT (e.g
    user123-850104-9/3/2025)
19    COUNT(DISTINCT v.CUSTOMER_EXID || '-' || v.CONTENT_ID || '-' || substr(v.STARTED_AT, 1,
    instr(v.STARTED_AT, ' ')-1)) AS unique_views
20
21 -- Data comes from the viewership table (all the watch events)
22 FROM viewership v
23
24 -- Join with the catalog table so each CONTENT ID can be linked to its external id
25 LEFT JOIN catalog c
26     ON v.CONTENT_ID = c.vod_dve_id
27     -- LEFT JOIN ensures we still keep the view even if the catalog row is missing (it becomes 'Other')
28
29 -- Group results so we get one row per source
30 GROUP BY source
31
32 -- Order results so the brands with the most views appear first
33 ORDER BY unique_views DESC;
```

Task 1: Total Views by Source (Brand)

```

1 -- Task 2: Average progress per video
2
3 -- Select the video id and title, and the average progress ratio for each video, and source
4 SELECT
5     v.CONTENT_ID,
6     -- Classify external_id prefixes to readable source names
7     CASE
8         WHEN c.external_id LIKE 'PGA%' THEN 'AJA'           -- If it starts with PGA → Al Jazeera Arabic
9         WHEN c.external_id LIKE 'PGD%' THEN 'AJD'           -- If it starts with PGD → Al Jazeera Documentary
10        WHEN c.external_id LIKE 'PGV%' THEN 'Atheer'         -- If it starts with PGV → Atheer (Podcasts)
11        WHEN c.external_id LIKE 'PGL%' THEN 'LIVE'           -- If it starts with PGL → Live channels
12        WHEN c.external_id LIKE 'PGC%' THEN 'Originals'      -- If it starts with PGC → Originals / Commissions
13        ELSE 'Other'                                         -- Everything else or missing external_id
14    END AS source, -- name it as source
15
16    -- Use the title from the catalog if available; otherwise use the CONTENT_ID as text
17    --coalesce "take the first non-null value"
18    -- max() because SQL requires an aggregate function when grouping
19    COALESCE(MAX(c.title), CAST(v.CONTENT_ID AS TEXT)) AS title,
20
21    -- Count how many watch events contributed to this average (helps spot small-sample noise)
22    COUNT(*) AS events_count,
23
24    -- Give video's duration (in sec). If multiple values exist, just take the maximum.
25    MAX(v.CONTENT_DURATION) AS content_duration_sec,
26
27    -- Calculate the average progress ratio (Ratio = PROGRESS / CONTENT_DURATION)
28    -- PROGRESS = seconds watched
29    -- CONTENT_DURATION = total video length in seconds
30    -- MIN(100.0, ...) ensures the ratio never goes above 100.0 (100%) (e.g if PROGRESS=900 and
    DURATION=600 → 900*100/600=150.0 → capped to 100.0)
31    AVG(
32        CASE
33            WHEN v.CONTENT_DURATION IS NULL OR v.CONTENT_DURATION = 0 THEN NULL -- Skip rows with
    missing/zero duration
34            ELSE MIN(100.0, (v.PROGRESS * 100.0) / v.CONTENT_DURATION)
35        END
36    ) AS avg_progress_ratio
37
38 -- Data comes from the viewership table (all watch events)
39 FROM viewership v
40
41 -- Join with the catalog table so we can display the title
42 LEFT JOIN catalog c
43     ON v.CONTENT_ID = c.vod_dve_id
44
45 -- Group results so we get one row per video
46 GROUP BY v.CONTENT_ID
47
48 -- Order results so the videos with the highest average progress appear first
49 ORDER BY avg_progress_ratio DESC;
50

```

Task 2: Average Progress per Video

```

1 -- Task 4: Most-watched series (unique viewers), with brand classification
2
3 SELECT
4     -- Classify external_id prefixes to readable source names
5     CASE
6         WHEN c.external_id LIKE 'PGA%' THEN 'AJA'           -- PGA → Al Jazeera Arabic
7         WHEN c.external_id LIKE 'PGD%' THEN 'AJD'           -- PGD → Al Jazeera Documentary
8         WHEN c.external_id LIKE 'PGV%' THEN 'Atheer'         -- PGV → Ather Podcasts
9         WHEN c.external_id LIKE 'PGL%' THEN 'LIVE'           -- PGL → Live channels
10        WHEN c.external_id LIKE 'PGC%' THEN 'Originals'      -- PGC → Originals / Commissions
11        ELSE 'Other'                                         -- Everything else
12    END AS source,
13
14    -- Series information
15    v.SERIES_ID,
16
17    -- coalesce replaces null values with "Unknown Series"
18    COALESCE(v.SERIES_TITLE, 'Unknown Series') AS series_title,
19
20    -- Count unique viewers (one user per series)
21    COUNT(DISTINCT v.CUSTOMER_EXID) AS unique_viewers,
22
23    -- Total number of view events (all plays, including repeats)
24    COUNT(*) AS total_views
25
26 FROM viewership v
27
28 -- Join to catalog to pull external_id (needed for source classification)
29 LEFT JOIN catalog c
30     ON v.CONTENT_ID = c.vod_dve_id
31
32 -- Keep only rows where SERIES ID is available
33 WHERE v.SERIES_ID IS NOT NULL
34
35 -- Group by source + series
36 GROUP BY source, v.SERIES_ID, v.SERIES_TITLE
37
38 -- Order by unique viewers (biggest series first)
39 ORDER BY unique_viewers DESC;
40

```

Task 3: Average Completion Rate

```

1  -- Task 3: Average completion rate per source (brand)
2
3  -- Select each source and calculate its completion rate
4  SELECT
5      -- Classify external_id prefixes to readable source names
6      CASE
7          WHEN c.external_id LIKE 'PGA%' THEN 'AJA'           -- If it starts with PGA → Al Jazeera Arabic
8          WHEN c.external_id LIKE 'PGD%' THEN 'AJD'           -- If it starts with PGD → Al Jazeera Documentary
9          WHEN c.external_id LIKE 'PGV%' THEN 'Atheer'         -- If it starts with PGV → Atheer (Podcasts)
10         WHEN c.external_id LIKE 'PGL%' THEN 'LIVE'           -- If it starts with PGL → Live channels
11         WHEN c.external_id LIKE 'PGC%' THEN 'Originals'       -- If it starts with PGC → Originals / Commissions
12         ELSE 'Other'                                           -- Everything else or missing external_id
13     END AS source, -- name it as source
14
15     -- Total number of valid watch events
16     COUNT(*) AS events_count,
17
18     -- Events reached completion (watched >= 90% of duration)
19     SUM(
20         CASE
21             WHEN v.CONTENT_DURATION > 0
22                 AND (v.PROGRESS * 100.0) / v.CONTENT_DURATION >= 90.0 THEN 1
23             ELSE 0
24         END
25     ) AS completed_events,
26
27     -- Completion rate (%) = completed_events / total_events * 100
28     100.0 * SUM(
29         CASE
30             WHEN v.CONTENT_DURATION > 0
31                 AND (v.PROGRESS * 100.0) / v.CONTENT_DURATION >= 90.0 THEN 1
32             ELSE 0
33         END
34     ) / COUNT(*) AS completion_rate_pct
35
36 -- Data comes from the viewership table
37 FROM viewership v
38
39 -- Join to catalog to classify each video into a brand/source
40 LEFT JOIN catalog c
41     ON v.CONTENT_ID = c.vod_dve_id
42
43 -- Only keep rows where duration is available and > 0 to avoid errors
44 WHERE v.CONTENT_DURATION IS NOT NULL
45     AND v.CONTENT_DURATION > 0
46
47 -- Group results so we get one row per source
48 GROUP BY source
49
50 -- Order brands by completion rate (highest first)
51 ORDER BY completion_rate_pct DESC;
52

```

Task 4: Most-watched series (unique viewers)

```

1 -- Task 5: New vs. Old content by source (simple version)
2
3 SELECT
4   -- Classify as New vs Old based only on year/month from created at
5   -- Year = 2025 (found via pattern '/2025 ') AND month in (7,8,9) → "New"
6   CASE
7     WHEN c.created_at LIKE '%/2025 %'
8       AND CAST(substr(c.created_at, 1, instr(c.created_at, '/') - 1) AS INTEGER) IN (7,8,9)
9       THEN 'New' -- if the month is July, August, or September, we tag the content as New
10    ELSE 'Old'
11  END AS content_age,
12
13  -- Source classification
14  CASE
15    WHEN c.external_id LIKE 'PGA%' THEN 'AJA' -- PGA → Al Jazeera Arabic
16    WHEN c.external_id LIKE 'PGD%' THEN 'AJD' -- PGD → Al Jazeera Documentary
17    WHEN c.external_id LIKE 'PGV%' THEN 'Atheer' -- PGV → Ather Podcasts
18    WHEN c.external_id LIKE 'PGL%' THEN 'LIVE' -- PGL → Live channels
19    WHEN c.external_id LIKE 'PGC%' THEN 'Originals' -- PGC → Originals / Commissions
20    ELSE 'Other' -- Everything else
21  END AS source,
22
23  -- Count unique viewers
24  COUNT(DISTINCT v.CUSTOMER_EXID) AS unique_viewers,
25
26  -- Total number of view events (all plays, including repeats)
27  COUNT(*) AS total_views
28
29 FROM viewership v
30 LEFT JOIN catalog c
31   ON v.CONTENT_ID = c.vod_dve_id
32
33 -- Only include rows where we have created_at available
34 WHERE c.created_at IS NOT NULL
35
36 -- Group by age + source
37 GROUP BY content_age, source
38
39 -- Order by new/old then biggest audiences
40 ORDER BY content_age DESC, unique_viewers DESC;
41

```

Task 5: New vs. Old Content

```

1 -- Task 6: Repeat viewers (number and share of users)
2
3 -- Build a temporary table of (user, calendar day) pairs
4 WITH user_days AS (
5     SELECT
6         v.CUSTOMER_EXID,
7
8         -- Extract just the calendar date part from STARTED AT (before the space between the date and the time)
9         CASE
10             WHEN instr(v.STARTED_AT, ' ') > 0
11             THEN substr(v.STARTED_AT, 1, instr(v.STARTED_AT, ' ') - 1) -- extracts a piece of text, begin at
12 the first character , stop before the space (e.g., '9/3/2025')
13             ELSE v.STARTED_AT -- fallback if no space is present
14         END AS view_date,
15         v.CONTENT_ID
16     FROM viewership v
17     -- One row per (user, day, content) so we can later tag each user activity to a source
18     GROUP BY v.CUSTOMER_EXID, view_date, v.CONTENT_ID
19 ),
20
21 -- Count distinct days per user (per content source)
22 user_day_counts AS (
23     SELECT
24         ud.CUSTOMER_EXID,
25         COUNT(DISTINCT ud.view_date) AS active_days, -- number of different days this user watched (within
26 source)
27         c.external_id
28     FROM user_days ud -- accessing the temporary table create usr_days
29     LEFT JOIN catalog c
30     ON ud.CONTENT_ID = c.vod_dve_id
31     GROUP BY ud.CUSTOMER_EXID, c.external_id
32 ),
33
34 user_sources AS (
35     -- classify users as repeat vs one-time, and tag to source
36     SELECT
37         CUSTOMER_EXID,
38         CASE
39             WHEN external_id LIKE 'PGA%' THEN 'AJA' -- PGA → Al Jazeera Arabic
40             WHEN external_id LIKE 'PGD%' THEN 'AJD' -- PGD → Al Jazeera Documentary
41             WHEN external_id LIKE 'PGV%' THEN 'Atheer' -- PGV → Ather Podcasts
42             WHEN external_id LIKE 'PGL%' THEN 'LIVE' -- PGL → Live channels
43             WHEN external_id LIKE 'PGC%' THEN 'Originals' -- PGC → Originals / Commissions
44             ELSE 'Other' -- Everything else
45         END AS source,
46         CASE WHEN active_days > 1 THEN 1 ELSE 0 END AS is_repeat -- users who came back on a different day
47     FROM user_day_counts
48 )
49
50 -- Calculate total users and repeat users per sources
51 SELECT
52     source,
53     COUNT(DISTINCT CUSTOMER_EXID) AS total_users, -- unique users for this source
54
55     -- Repeat users = users who watched on more than 1 calendar day
56     SUM(is_repeat) AS repeat_users,
57
58     -- Share of repeat users as a percentage of all users
59     ROUND(100.0 * SUM(is_repeat) / COUNT(DISTINCT CUSTOMER_EXID), 2) AS repeat_share_pct
60 FROM user_sources -- accessing the table create user sources
61 GROUP BY source
62 ORDER BY repeat_share_pct DESC;

```

Task 6: Repeat Viewers (overall and by source)

```

1 -- Task 7: Top-viewed content list (episodes)
2
3 SELECT
4     CASE
5         WHEN external_id LIKE 'PGA%' THEN 'AJA'           -- PGA → Al Jazeera Arabic
6         WHEN external_id LIKE 'PGD%' THEN 'AJD'           -- PGD → Al Jazeera Documentary
7         WHEN external_id LIKE 'PGV%' THEN 'Atheer'         -- PGV → Ather Podcasts
8         WHEN external_id LIKE 'PGL%' THEN 'LIVE'           -- PGL → Live channels
9         WHEN external_id LIKE 'PGC%' THEN 'Originals'       -- PGC → Originals / Commissions
10        ELSE 'Other'                                         -- Everything else
11    END AS source,
12    c.title,           -- episode title
13    c.series title,    -- series name (if available)
14    c.thumbnail_url,   -- thumbnail image link for title
15    COUNT(DISTINCT v.CUSTOMER_EXID || '-' || v.CONTENT_ID || '-' ||
16           substr(v.STARTED AT, 1, instr(v.STARTED AT, '-')-1)) AS unique_views -- unique view = (user +
content + calendar day)
17
18 FROM viewership v
19 LEFT JOIN catalog c
20     ON v.CONTENT ID = c.vod dve id
21
22 WHERE c.title IS NOT NULL -- exclude cases with no title (null)
23        AND TRIM(c.title) <> '' -- exclude rows where the title is just an empty string
24
25 GROUP BY c.title, c.series title, c.thumbnail url -- to get one row per episode.
26
27 ORDER BY unique_views DESC -- find the most watched
28
29 LIMIT 20; -- show top 20 episodes
30

```

Task 7: Top-Viewed Content List (Episodes with Thumbnails)